mhaynes@eas.gatech.edu

Towards a Constrained Understanding of Europa's Subsurface Driven by Magnetic Induction Sounding

C. Michael Haynes

EAS 6134

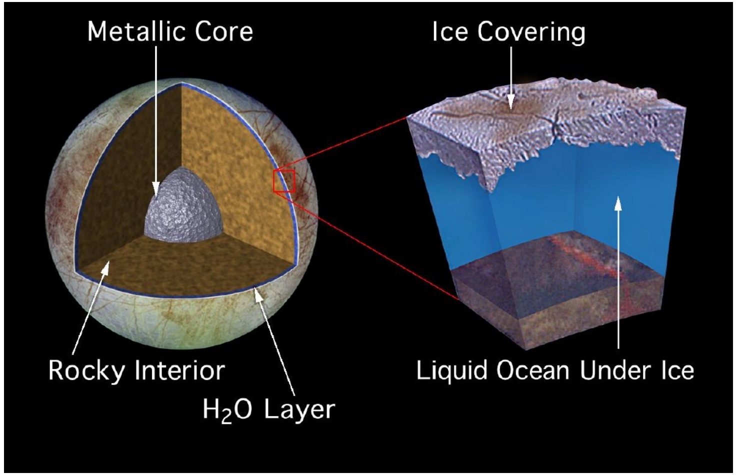
Georgia Institute of Technology

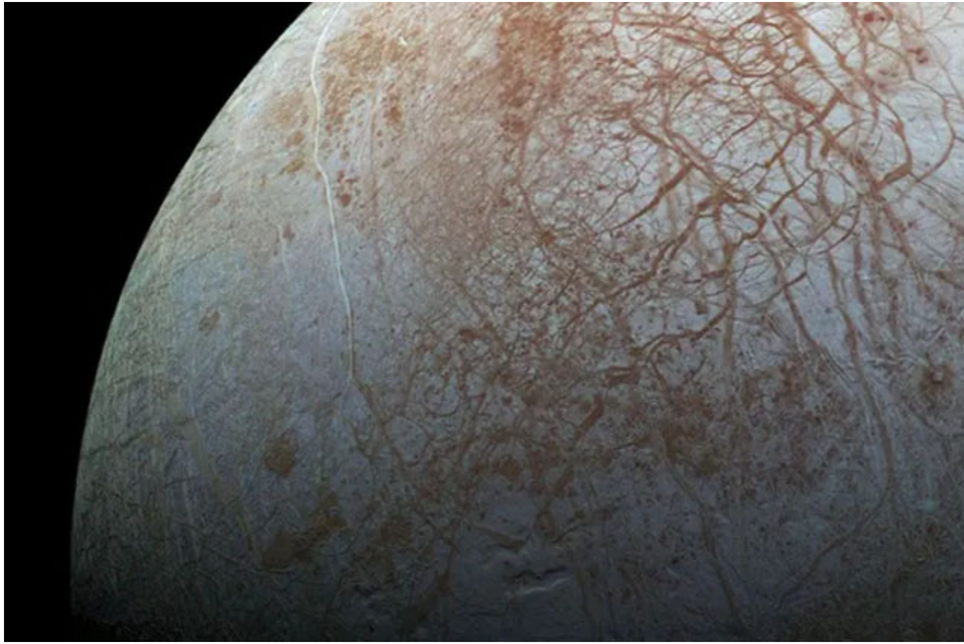
NASA

mhaynes@eas.gatech.edu

Europa: Background

Europa: $R_E = 1561\ \mathrm{km}$



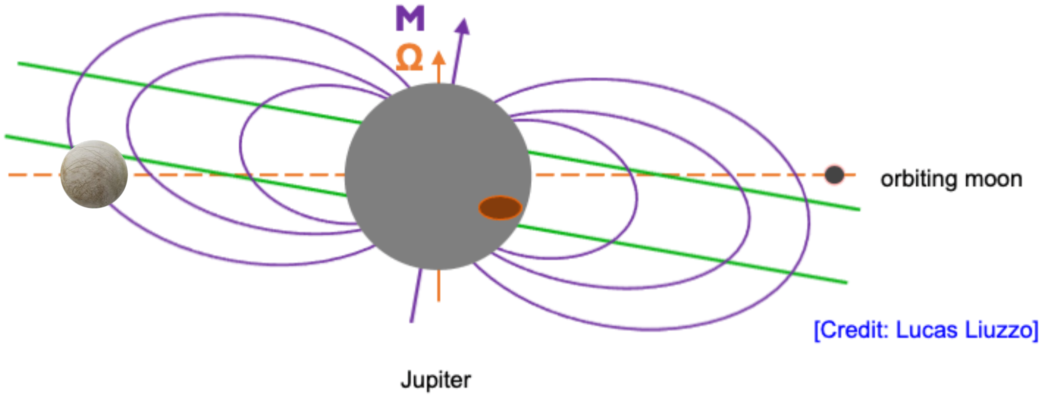


- Embedded in Jupiter's magnetosphere
- Target of the upcoming JUICE and Europa Clipper missions
- Possesses many habitable qualities (water, active geochemistry)

mhaynes@eas.gatech.edu

Europa: Background

M: Magnetic Axis
Ω: Rotation Axis
Plasma Sheet



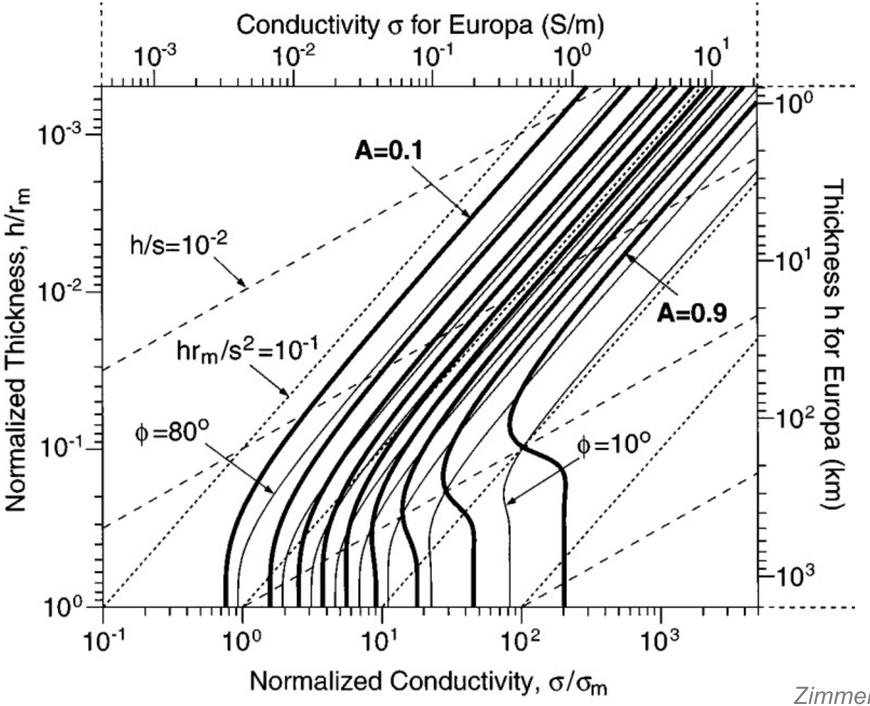
[Credit: Lucas Liuzzo]

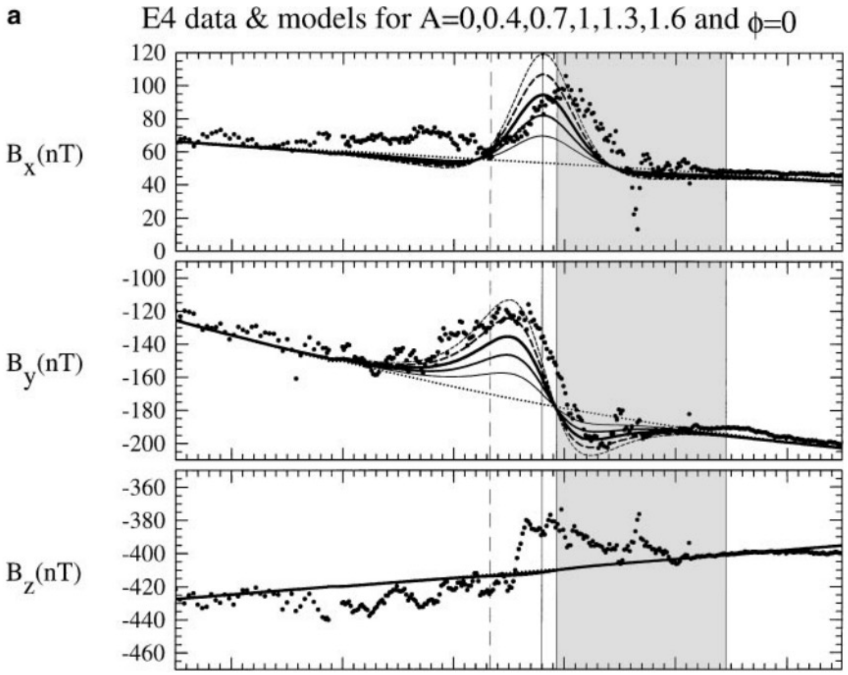
Analogs:

- Offset between Jupiter’s spin and rotation axes
- ➔ time-varying magnetospheric field **B** at Europa’s orbit
- Drives induction in conducting layers of the moon (salty water)
- Led to the discovery of Europa’s ocean! (Kivelson et al., 2000)

mhaynes@eas.gatech.edu

Science Question: Where Is the Ocean?





Zimmer et al. (2000)

mhaynes@eas.gatech.edu

Forward Operator: Induction Physics

- Dominant harmonic: synodic rotation period of Jupiter

- Induction described by: $\nabla^2 \mathbf{B} = \mu_0 \sigma \, \partial_t \mathbf{B} \quad ; \quad \text{if } \sigma \longrightarrow 0 \quad \Longrightarrow \quad \nabla^2 \mathbf{B} = 0$

➔ spherical harmonic decomposition assuming spherical symmetry isotropic shape

- For a prescribed subsurface conductivity profile, the inductive response is:

$$\mathbf{B}_{\text{ind}} = \frac{\mu_0}{4\pi} \left(3(\mathbf{r} \cdot \mathbf{M}_{\text{ind}})\mathbf{r} - \mathbf{r}^2 \mathbf{M}_{\text{ind}} \right) / r^5$$

Vance et al. (2021)

mhaynes@eas.gatech.edu

Forward Operator: Induction Physics

- Dominant harmonic: synodic rotation period of Jupiter

- Induction described by: $\nabla^2 \mathbf{B} = \mu_0 \sigma \, \partial_t \mathbf{B} \quad ; \quad \text{if } \sigma \longrightarrow 0 \quad \Longrightarrow \quad \nabla^2 \mathbf{B} = 0$

➔ spherical harmonic decomposition assuming spherical symmetry isotropic shape

- For a prescribed subsurface conductivity profile, the inductive response is:

$$\mathbf{B}_{\text{ind}} = \frac{\mu_0}{4\pi} \left(3(\mathbf{r} \cdot \mathbf{M}_{\text{ind}})\mathbf{r} - \mathbf{r}^2 \mathbf{M}_{\text{ind}} \right) / r^5$$

Vance et al. (2021)

$$\boxed{\mathbf{M}_{\text{ind}} = -A\, e^{-i(\omega t - \phi)} \frac{2\pi R_E^3}{\mu_0} \mathbf{B}_{h,0}}$$

mhaynes@eas.gatech.edu

Forward Operator: Induction Physics

$$\mathbf{M}_{\mathrm{ind}} = -A\, e^{-i(\omega t - \phi)}\, \frac{2\pi R_E^3}{\mu_0}\, \mathbf{B}_{h,0}$$

$$Ae^{i\phi} = \left(\frac{r_0}{R_E}\right)^3 \frac{C J_{\frac{5}{2}}(r_0\, k) - J_{-\frac{5}{2}}(r_0\, k)}{C J_{\frac{1}{2}}(r_0\, k) - J_{-\frac{1}{2}}(r_0\, k)}$$

Amplitude, phase lag dependence

$$C \equiv \frac{r_1 k J_{-\frac{5}{2}}(r_1\, k)}{3 J_{\frac{3}{2}}(r_1\, k) - r_1 k J_{\frac{1}{2}}(r_1\, k)}$$

coefficient dependent on ocean depth

$$k = (1-i)\sqrt{\frac{\mu_0 \sigma_o\, \omega}{2}} = (1-i)\sqrt{\frac{\pi \mu_0 \sigma_o}{T_s}}$$

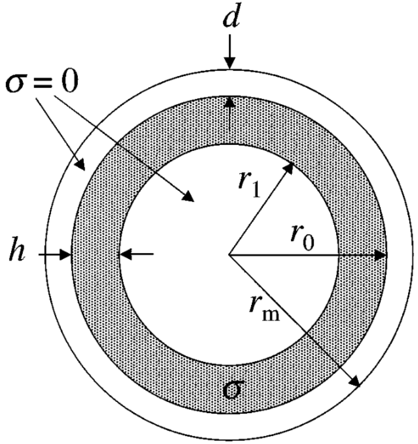
complex wavenumber dependent on conductivity value

mhaynes@eas.gatech.edu

Strategy: Model Assumptions & Discretization

$$\mathbf{M}_{\text{ind}} = -A\, e^{-i(\omega t-\phi)}\, \frac{2\pi R_E^3}{\mu_0}\, \mathbf{B}_{h,0}$$

(Zimmer et al., 2000)



- ➔ A, ϕ are amplitude and phase lag of induced field signal
- ➔ Assume that Europa's subsurface is discretized into 3 layers:
  - ➔ Uniform conductivity!
- ➔ A, ϕ dependent on subsurface ocean thickness, depth, σ

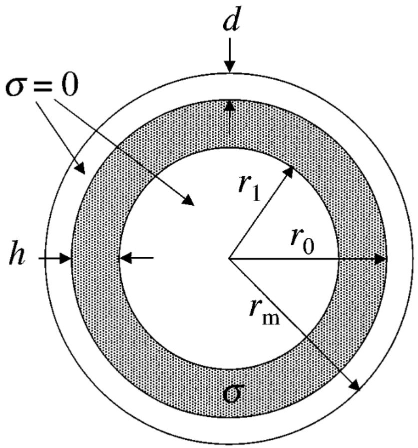
mhaynes@eas.gatech.edu

Inversion Strategy

- Our assumed model of Europa's subsurface is:

$$\mathbf{m} = [r_1, r_0, \sigma]$$

(Zimmer et al., 2000)



mhaynes@eas.gatech.edu

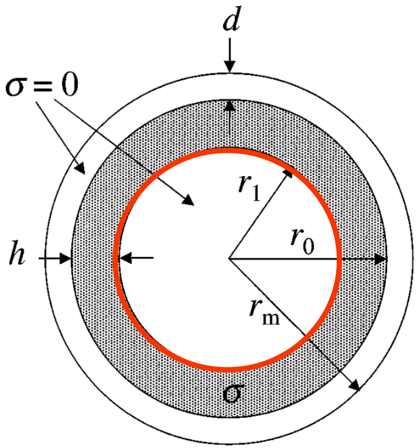
Inversion Strategy

- Our assumed model of Europa's subsurface is:

$$\mathbf{m} = [\boxed{r_1}, r_0, \sigma]$$

Inner radius

(Zimmer et al., 2000)



mhaynes@eas.gatech.edu

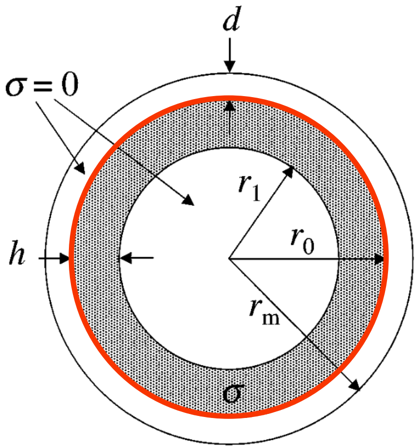
Inversion Strategy

- Our assumed model of Europa's subsurface is:

$$\mathbf{m} = [r_1, \boxed{r_0}, \sigma]$$

(Zimmer et al., 2000)

Outer radius



mhaynes@eas.gatech.edu

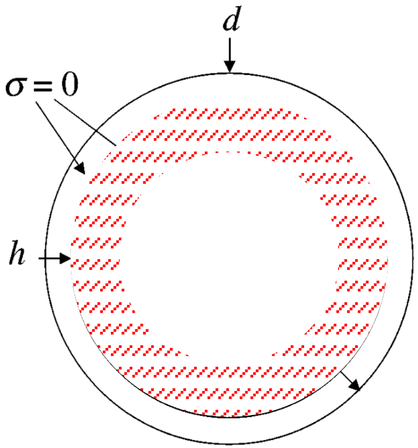
Inversion Strategy

- Our assumed model of Europa's subsurface is:

$$\mathbf{m} = [r_1, r_0, \boxed{\sigma}]$$

Conductivity

(Zimmer et al., 2000)



mhaynes@eas.gatech.edu

Inversion Strategy: Deterministic

- For our nonlinear deterministic approaches, the objective function is:

$$f(\mathbf{m}) = \sum_{i=1}^{M} \left(\frac{G(\mathbf{m})_i - d_i}{s_i} \right)^2$$

- We employ the Gauss-Newton method:

$$\left(\mathbb{J}(\mathbf{m})^{\mathrm{T}} \cdot \mathbb{J}(\mathbf{m})\right) \cdot \Delta\mathbf{m} = -\mathbb{J}(\mathbf{m})^{\mathrm{T}} \cdot \mathbf{F}(\mathbf{m})$$

- And the Levenberg-Marquardt Method (modified version: Trust Region Reflective)

$$\left(\mathbb{J}(\mathbf{m})^{\mathrm{T}} \cdot \mathbb{J}(\mathbf{m}) + \lambda\mathbb{I}\right) \cdot \Delta\mathbf{m} = -\mathbb{J}(\mathbf{m})^{\mathrm{T}} \cdot \mathbf{F}(\mathbf{m})$$

mhaynes@eas.gatech.edu

Inversion Strategy: Stochastic

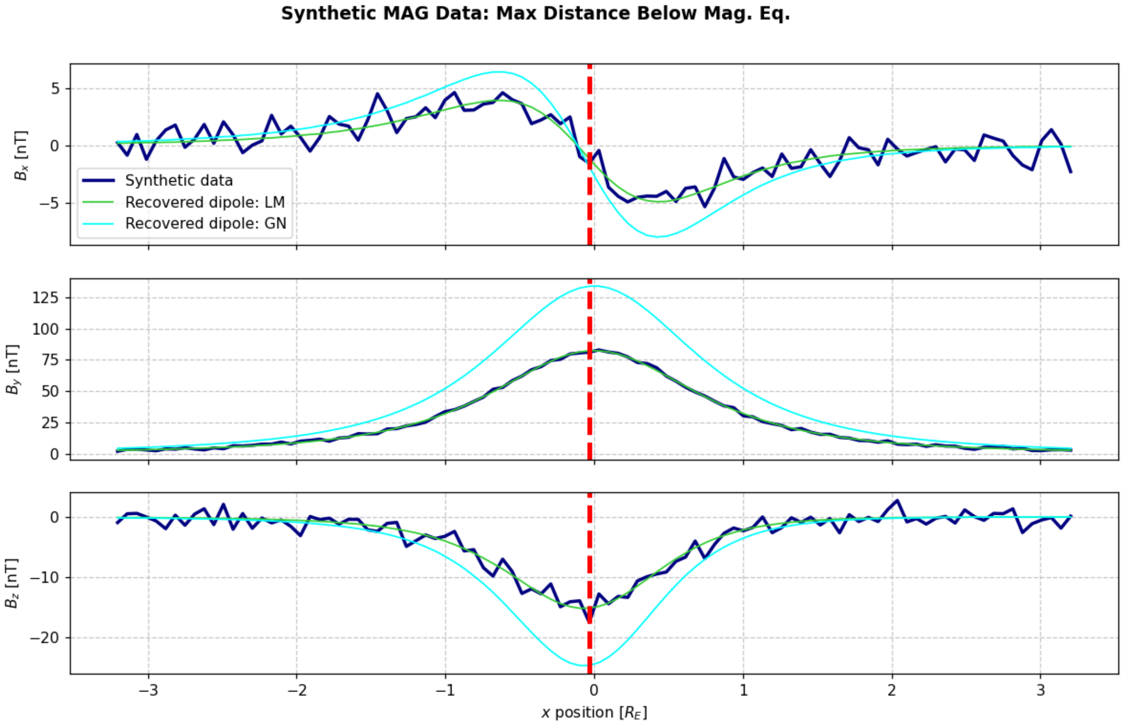
- Due to the high degree of non-uniqueness, we also implement the Metropolis-Hastings algorithm (Markov Chain Monte Carlo):

  - 800,000 iterations
  - implicit regularization (use $\log(\sigma)$ as 3rd parameter: accounts for higher sensitivity)
  - step sizes of [0.3, 0.16, 0.02]
  - burn-in: first 100,000 samples

mhaynes@eas.gatech.edu

Results: Synthetic Data

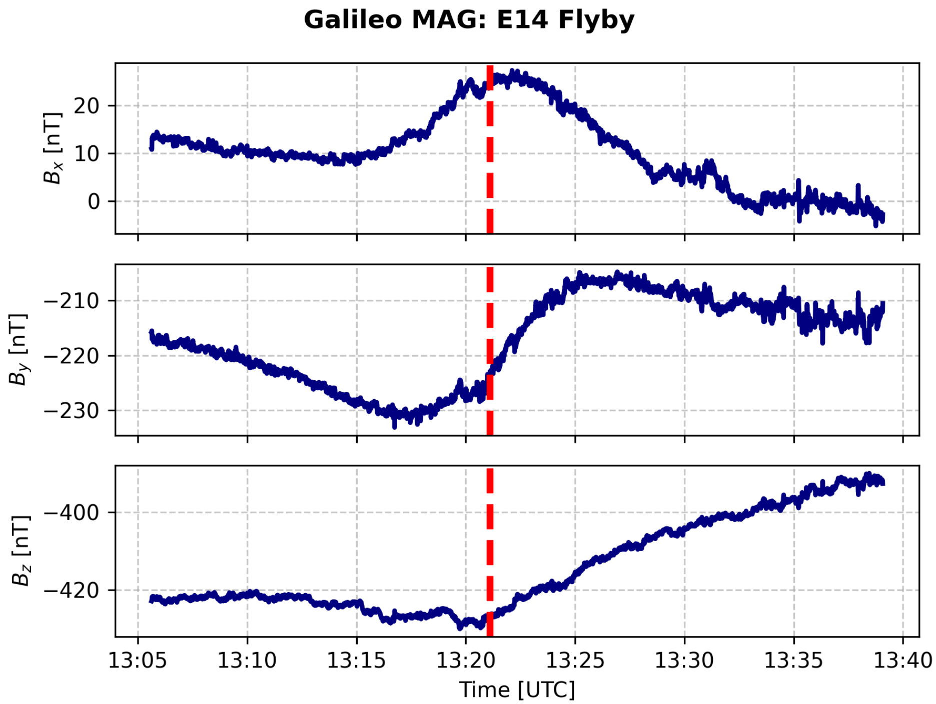
m_true = [1400 km, 1540km, 2.5 S/m]



$$\mathbf{m}_{\mathrm{G-N}} = [r_1 = 1309.04 \text{ km},\ r_0 = 1515.82 \text{ km},\ \sigma = 2.279 \text{ S/m}]$$
$$\mathbf{m}_{\mathrm{L-M}} = [r_1 = 1300.46 \text{ km},\ r_0 = 1350.41 \text{ km},\ \sigma = 2.69 \text{ S/m}]$$

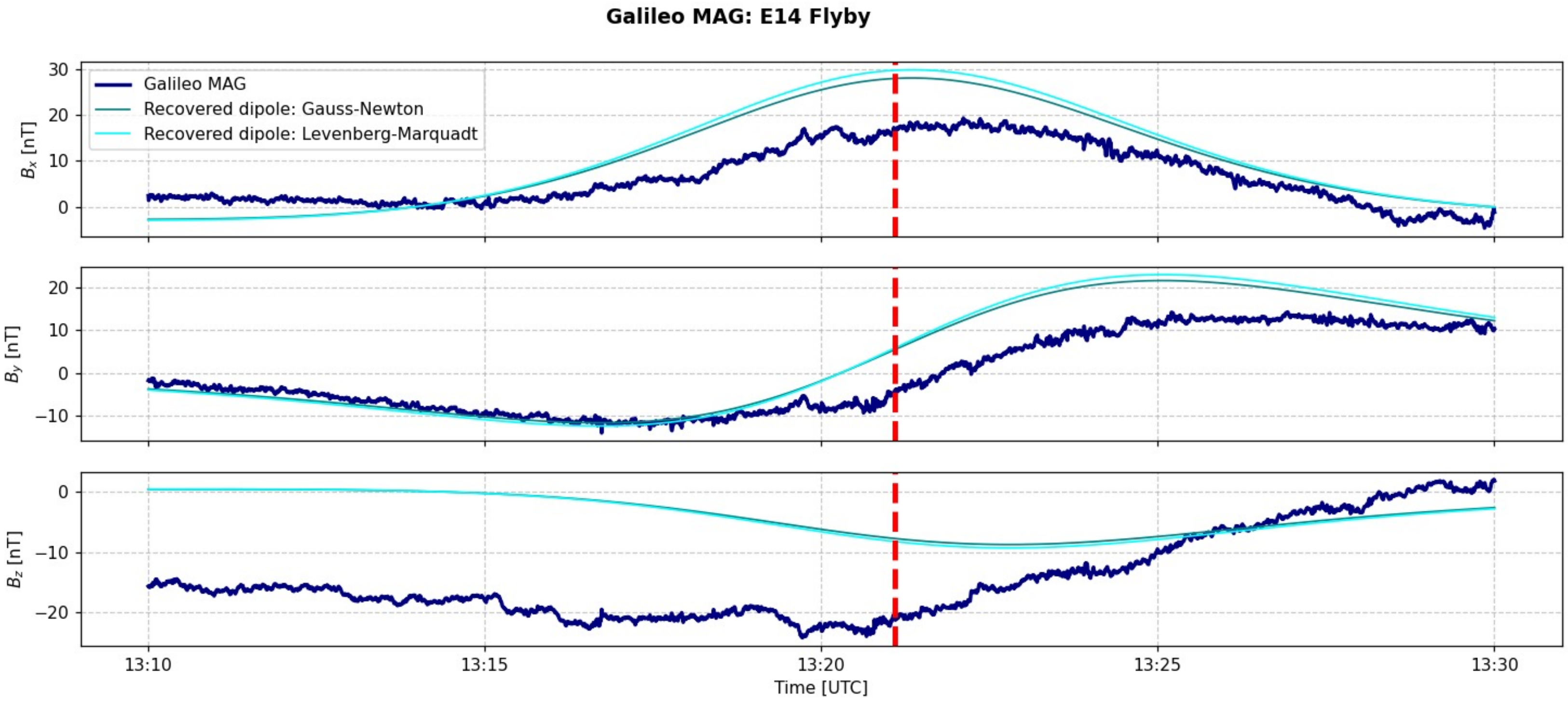
mhaynes@eas.gatech.edu

Results: Real Data



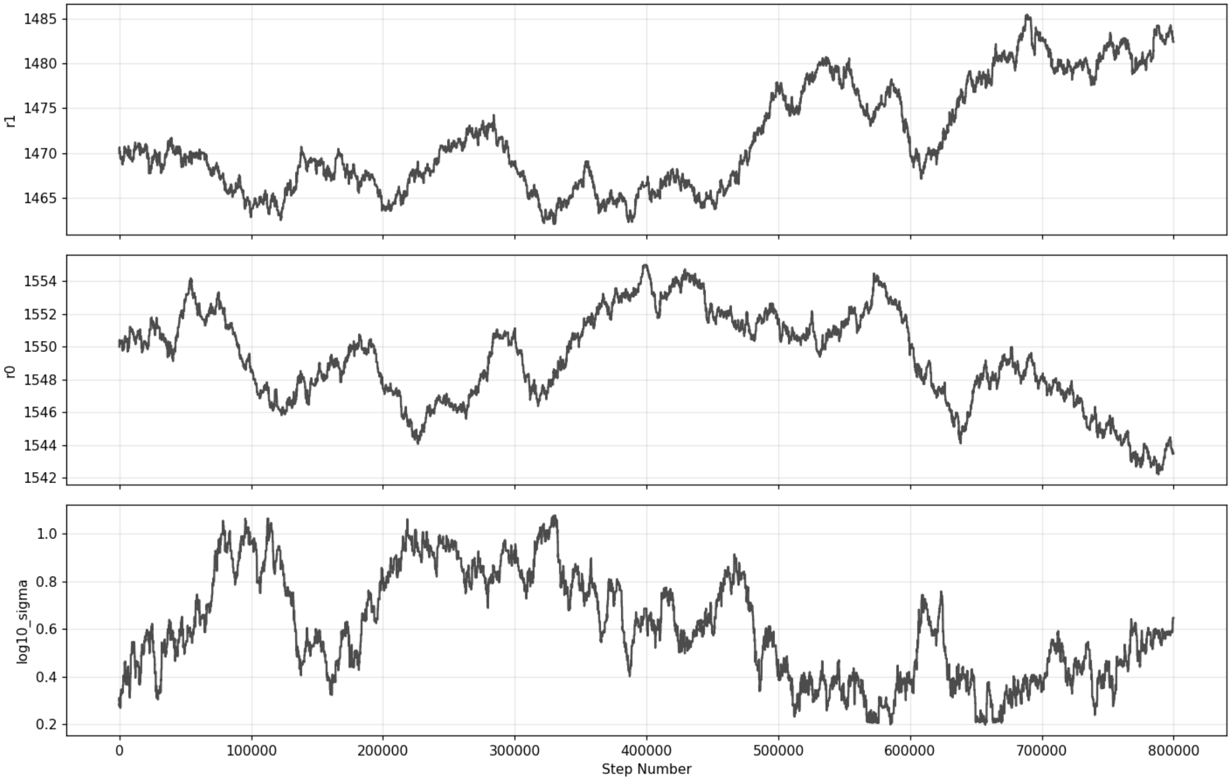
mhaynes@eas.gatech.edu

Results: Real Data



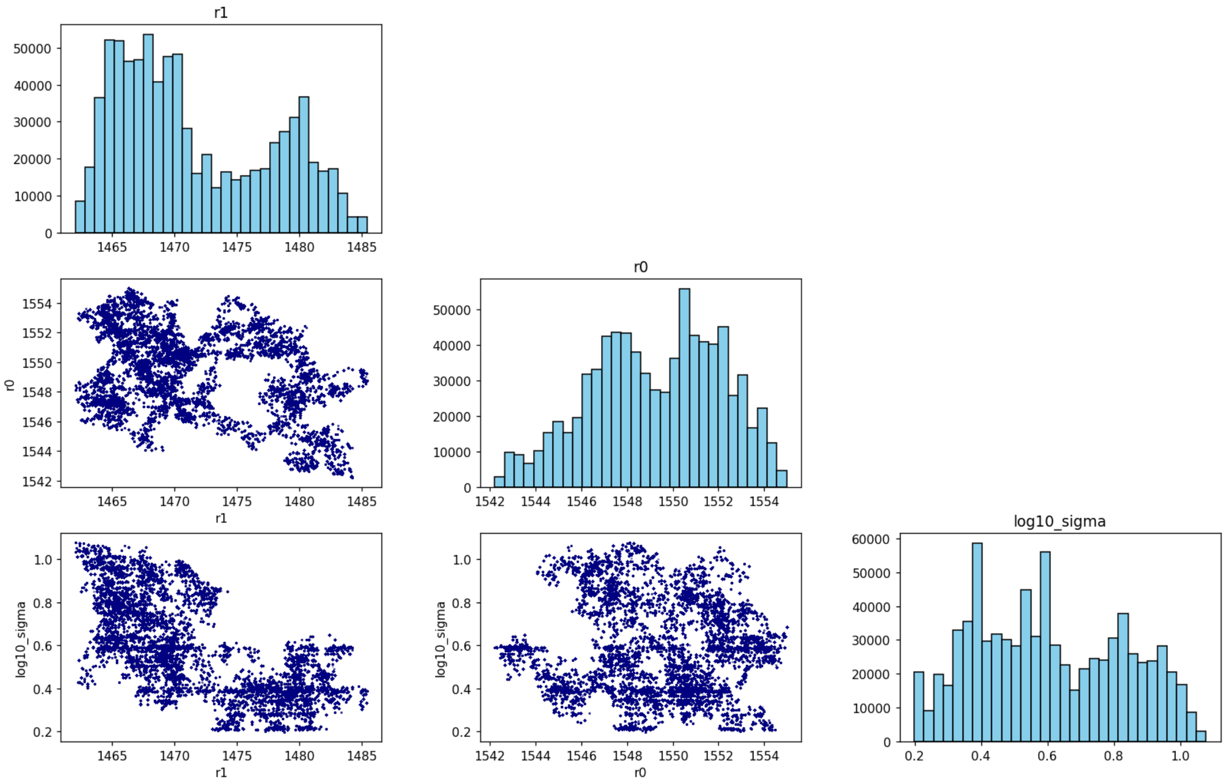
mhaynes@eas.gatech.edu

Statistical Results: Real Data



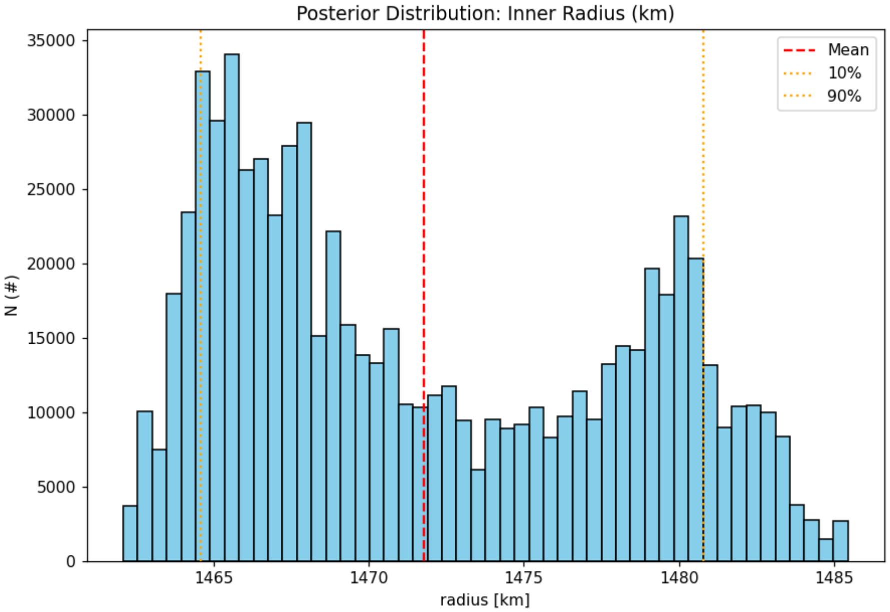
mhaynes@eas.gatech.edu

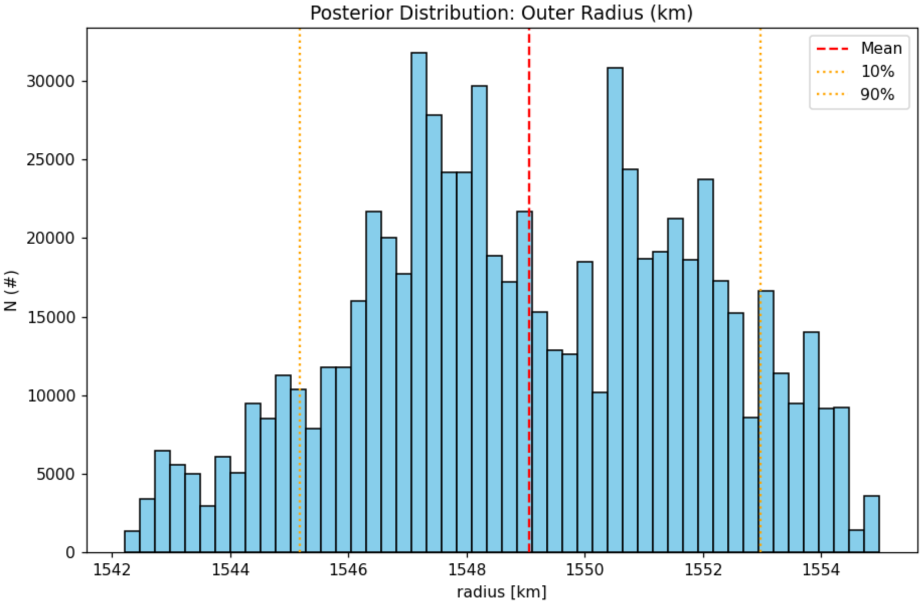
Statistical Results: Real Data



mhaynes@eas.gatech.edu

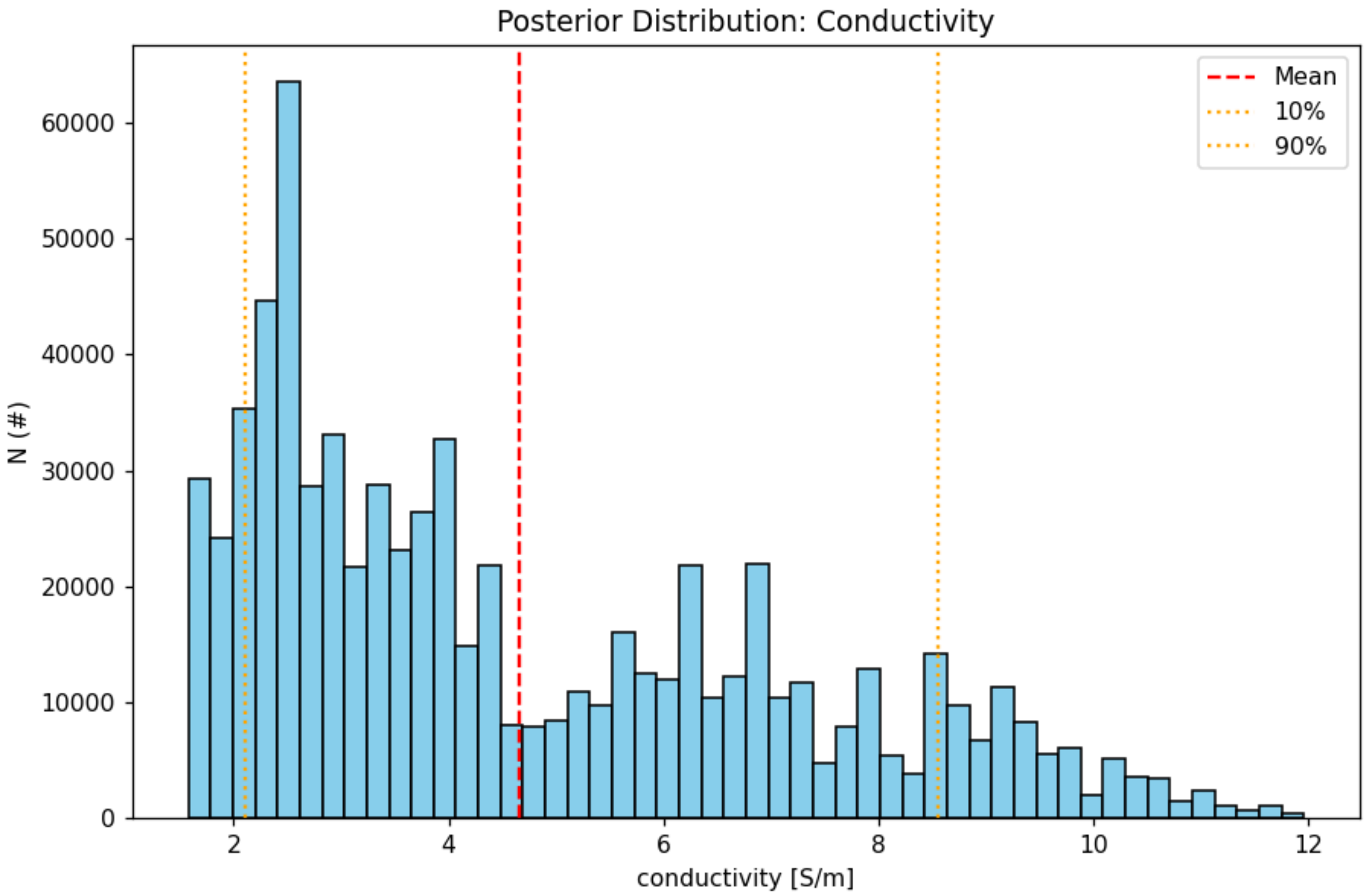
Statistical Results: Real Data





mhaynes@eas.gatech.edu

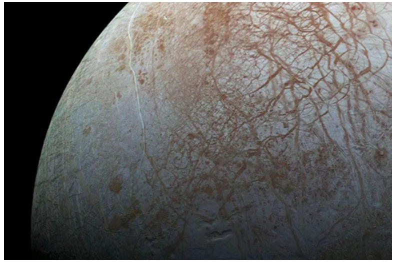
Statistical Results: Real Data

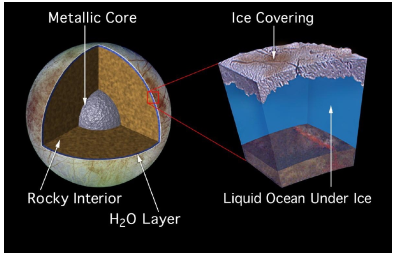


mhaynes@eas.gatech.edu

Summary: Magnetic Induction Sounding at Europa

Goal: Constrain the depth, thickness, & conductivity of Europa's ocean

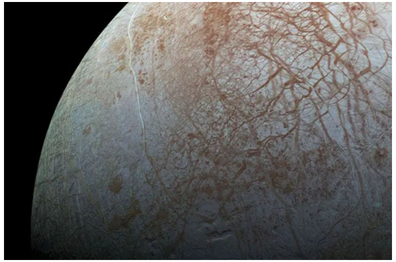


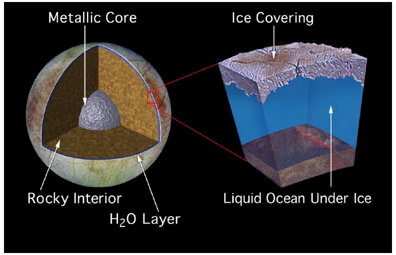


mhaynes@eas.gatech.edu

Summary: Magnetic Induction Sounding at Europa

<u>Goal</u>: Constrain the depth, thickness, & conductivity of Europa's ocean





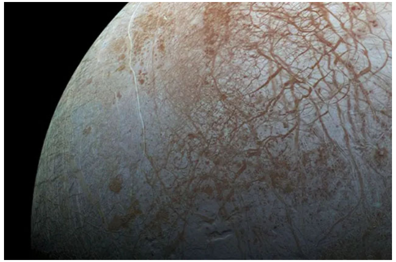
<u>Method</u>: Combine in-situ magnetometer observations, context-driven data selection, and nonlinear Inverse methods (G-N, L-M, MCMC)

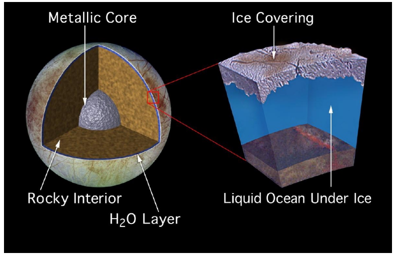
$$\left(\mathbb{J}(\mathbf{m})^{\mathrm{T}} \cdot \mathbb{J}(\mathbf{m})\right) \cdot \Delta\mathbf{m} = -\mathbb{J}(\mathbf{m})^{\mathrm{T}} \cdot \mathbf{F}(\mathbf{m})$$

mhaynes@eas.gatech.edu

Summary: Magnetic Induction Sounding at Europa

Goal: Constrain the depth, thickness, & conductivity of Europa's ocean

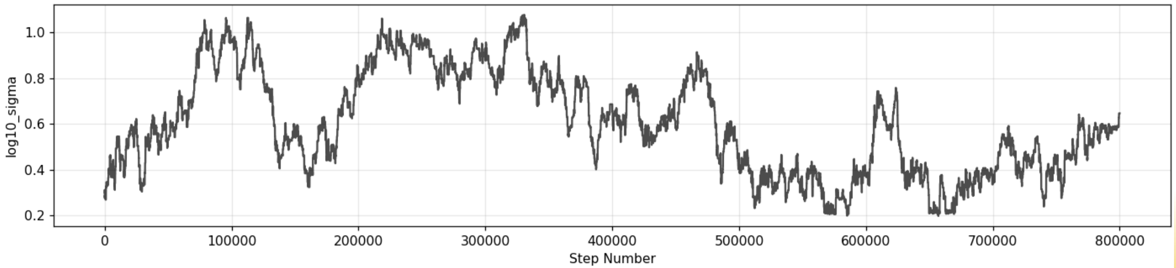


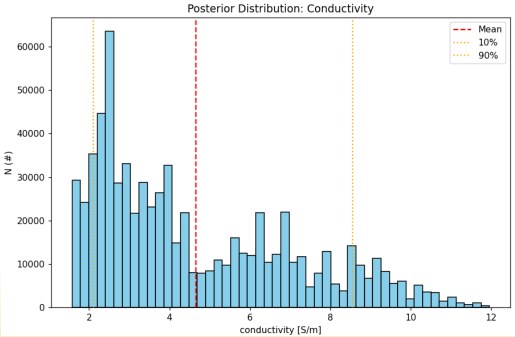


Method: Combine in-situ magnetometer observations, context-driven data selection, and nonlinear Inverse methods (G-N, L-M, MCMC)

$$\left(\mathbb{J}(\mathbf{m})^{\mathrm{T}} \cdot \mathbb{J}(\mathbf{m})\right) \cdot \Delta\mathbf{m} = -\mathbb{J}(\mathbf{m})^{\mathrm{T}} \cdot \mathbf{F}(\mathbf{m})$$

Results: This problem is extremely non-unique! However, Europa definitely has a salty, liquid water ocean, with a thickness of ~100 km, depth of 10-20km, and conductivity of 1-5 S/m





Thank you